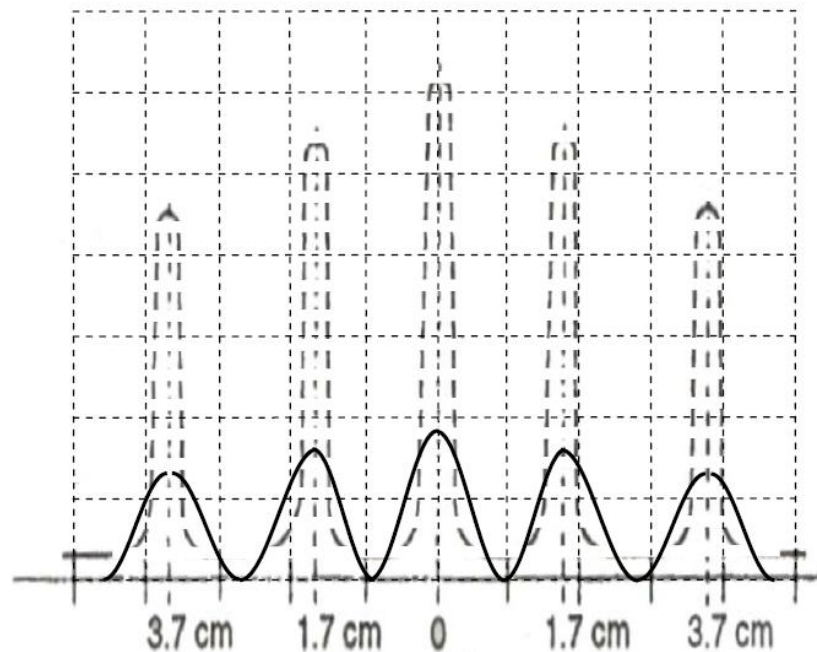


8	(a)	(i)	Wavelength = speed $\times$ time for 1 cycle $= 20 \times 0.60$ $= 12 \text{ cm}$ [A1]	1
		(ii)	Phase difference = $(0.20/0.60) \times 360^\circ = 120^\circ$ [A1] Or Phase difference = $(0.40/0.60) \times 360^\circ = 240^\circ$	1
		(iii)	Intensity $I \propto A^2$ [C1] $I_Q / I_P = A_Q^2 / A_P^2$ $= 2.0^2 / 3.0^2$ $= 0.44$ [A1]	1 1
		(iv)	Displacement = $1.00 - 3.00 = -2.00 \text{ mm}$ [A1]	1
	(b)	(i)	It means the two waves have <u>constant phase difference</u> . [A1]	1
		(ii)	P is the second minima from the central bright fringe, Path difference = $1.5\lambda = 1.5 \times 610$ $= 920 \text{ nm}$ [A1]	1
		(iii)	$\lambda = ax / D$ $x = (22 / 4) = 5.5. \text{ mm}$ [C1] $a = (610 \times 10^{-7})(2.7) / (5.5 \times 10^{-3})$ $= 3.0 \times 10^{-4} \text{ m}$ [A1]	1 1

		(iv)	When visible light of higher frequency is used, the wavelength becomes shorter. Since fringe separation is directly proportional to wavelength for the same slit separation and distance D, the fringe separation decreases. [B1]	1
		(v)	No change to fringe separation / fringe width / number of fringes [B1] Bright fringes are brighter, Dark fringes unchanged. [B1]	2
	(c)	(i)	Angular speed of grating $\omega = 2\pi/T = 2\pi/3.0 = 2.1 \text{ rad s}^{-1}$	1
		(ii)	When the turntable rotates, the second order of the diffraction pattern will be registered first by the detector, forming peak A. As the turntable continues rotating, the first order of the diffraction pattern (peak B) was detected and then the central maximum (peak C) was observed. Since the turntable rotates at constant angular speed, the other half of the peaks (D and E) appear at same separation away from the central maximum as peaks B and A respectively.	1 1
		(iii)	Time to shift from zeroth order to peak D through 1.7 cm on the screen is $1.7 \times 0.10 = 0.17 \text{ s}$ [C1] $\theta_1 = \omega t = (2\pi/3.0) (0.17) = 0.356 \text{ rad}$ $d \sin \theta_1 = n\lambda$ $\sin(0.356) = (1)(5.5 \times 10^5)(\lambda)$ [C1] $\lambda = 6.34 \times 10^{-7} \text{ m}$ [A1]	1 1 1

(iv)



Since the slit separation remain the same, the fringe separation remains the same. [B1]

1

However, the peaks are less intense (less contrast between the bright and dark fringes) and less sharp (broader and less defined) [B1]

1